

SatQuery AI Mission Report

Analysis type: cross_modal_analysis
Request ID: 81496f78-f1f2-470e-92b0-657cc4947325
Generated: 2026-09-05T12:58:35.933029Z
Status: success

Evidence-backed local remote-sensing analysis with an auditable execution trace.

Input Summary

Optical observation

File Id: 7f29db5f-ff09-438f-8438-e53c95bce706
Observation Role: optical
Auto Detected Modality: optical_rgb
Effective Modality: optical_rgb
Safe Name: optical_1.png
Format: png
Width: 256
Height: 256
Band Count: 3
Available Band Names: ['r', 'g', 'b']
Selected Visual Bands: ['r', 'g', 'b']
Band Selection Reason: color interpretation explicitly identifies red, green, and blue
Dtype: uint8
Crs: Unavailable
Is Georeferenced: False
Bounds: Unavailable

SAR observation

File Id: b92f0bac-984e-4f03-9a03-acf6b19ae9d0
Observation Role: sar
Auto Detected Modality: sar_preview
Effective Modality: sar_preview
Safe Name: sar_1.png
Format: png
Width: 256
Height: 256
Band Count: 1
Available Band Names: ['I']
Selected Visual Bands: ['I']

Band Selection Reason: single scientific band rendered as grayscale

Dtype: uint8

Crs: Unavailable

Is Georeferenced: False

Bounds: Unavailable

Compatibility

Compatible: True

Alignment Level: visual_only

Scientific Classification: UNVERIFIABLE

Same Dimensions: True

Same Crs: Unavailable

Same Transform: Unavailable

Same Resolution: Unavailable

Same Orientation: Unavailable

Nodata Compatible: Unavailable

Overlap Ratio: Unavailable

Resampling Required: False

Recommended Action: Same-size PNG/JPEG pairs can use disclosed qualitative native-source comparison. Provide verified CRS and affine transforms for quantitative fusion.

Query and Answer

Original query: Where do both modalities agree?

Detected task: cross_modal_analysis

Question category: cross_modal_agreement

Target concept: agreement

Qualitative optical-SAR analysis. Both observations have water candidate support in the lower-center, lower-left, lower-right, middle-center, middle-left, middle-right, upper-right image-relative sectors: optical dark/blue smooth optical appearance consistent with water; native SAR low-backscatter evidence consistent with water; radar shadow and smooth terrain remain alternatives. Both observations have structure candidate support in the lower-center, lower-left, lower-right, middle-center, middle-left, middle-right, upper-center, upper-left, upper-right image-relative sectors: optical visible edges and texture consistent with structural terrain; native SAR structural/high-return evidence consistent with built-up terrain; bright terrain and vegetation remain alternatives. Pixel alignment is assumed from the supplied pairing; geospatial co-registration cannot be independently verified. Locations refer to image sectors, not geographic directions.

Answer source: deterministic optical-SAR qualitative native evidence

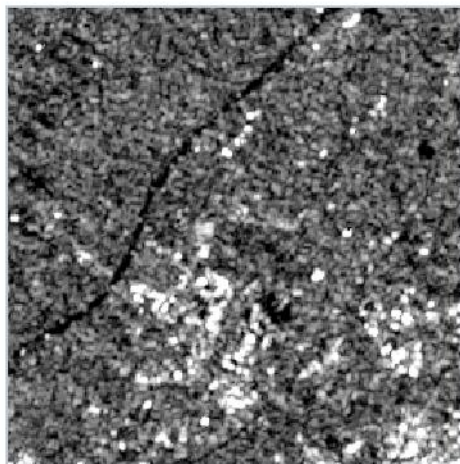
Evidence Products

Evidence products are model-generated or deterministic heuristic products as labelled; none are ground truth.

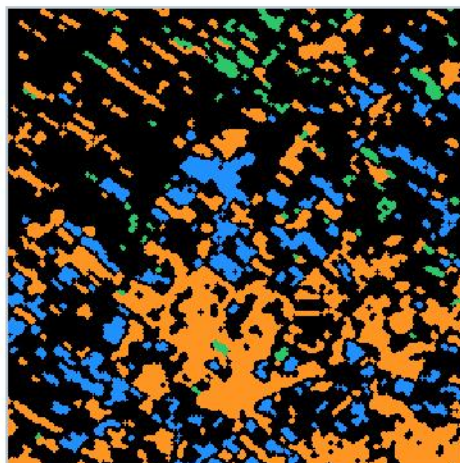
Optical source



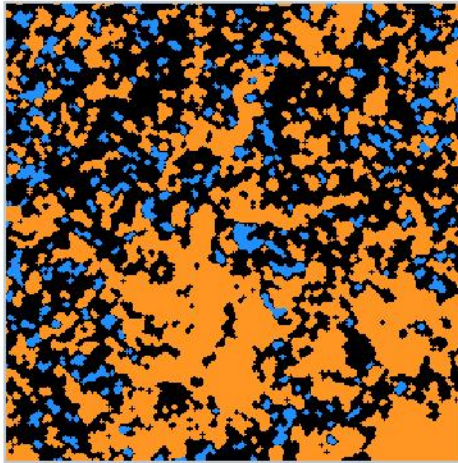
SAR source



Optical Evidence



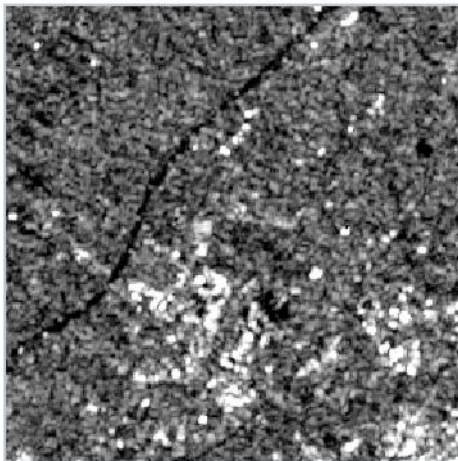
Sar Evidence



Optical-SAR · Optical



Optical-SAR · Sar



Statistics

Controlled Answer Statistics

Analysis Level: qualitative

Quantitative Metrics Available: False

Sve Scene Evidence

Status: success

Device: cpu

Runtime Ms: 6301

Caption Consistency: Unavailable

Vqa Consistency: Unavailable

Grounding Support: Unavailable

Confidence and Provenance

Confidence level: moderate

Calibrated score: Unavailable

Rationale: Both native sources were analyzed independently with conservative candidate rules; registration and semantic certainty remain unverified. This is not a calibrated probability.

Selected tools: input_validator, cross_modal_optical_sar_analyzer

Remote-Sensing Adaptation

Sve Model: SatQuery Vision Encoder v1

Sve Model Version: 1.0.0

Sve Backbone: OpenCLIP ViT-L-14

Sve Adaptation Dataset: BigEarthNet.txt

Sve Adapter Checksum Fingerprint: sha256:a99c0bf0fb44

Sve Device: cpu

Sve Runtime Ms: 6301

Sve Status: success

Scene prior · agricultural land: 0.260688

Scene prior · urban or built-up area: 0.255139

Scene prior · arable land: 0.243113

Scene prior · residential area: 0.242427

Scene prior · industrial area: 0.233264

SatQuery Vision Encoder v1 provides scene-level embedding evidence. It does not produce pixel-level segmentation, object grounding, calibrated probabilities, or ground truth.

Runtime: 6361 ms

Confidence values are reported only where they are real; heuristic semantic outputs are not calibrated probabilities.

Observable Execution Trace

1. optical_upload_received · success · 0 ms

{"size_bytes": 146114}

2. sar_upload_received · success · 0 ms

{"size_bytes": 46519}

3. optical_metadata_extraction · success · 0 ms

{"bands": 3, "georeferenced": false, "height": 256, "width": 256}

4. sar_metadata_extraction · success · 0 ms

{"bands": 1, "georeferenced": false, "height": 256, "width": 256}

5. modality_validation · success · 0 ms

```

{"optical_modality": "optical_or_multispectral", "optical_observation_id": "7f29db5f-ff09-438f-8438-e53c95bce706", "role_validation_status": "match", "sar_modality": "sar", "sar_observation_id": "b92f0bac-984e-4f03-9a03-acf6b19ae9d0"}

6. pair_compatibility_check · success · 0 ms
{"alignment_level": "visual_only", "reprojection_performed": false, "resampling_performed": false}

7. optical_preparation · success · 2 ms
{"max_analysis_dimension": 1024, "registration_performed": false}

8. sar_preparation · success · 1 ms
{"calibrated_backscatter_claimed": false, "log_transform": "none"}

9. optical_evidence_extraction · success · 9 ms
{"method": "visible_color_brightness_edges_local_texture"}

10. sar_evidence_extraction · success · 6 ms
{"method": "relative_intensity_and_local_heterogeneity"}

11. joint_evidence_fusion · success · 0 ms
{"method": "independent_image_sector_comparison"}

12. region_extraction · skipped · 0 ms
{"connectivity": 8, "minimum_region_pixels": 6, "reason": "No geospatial co-registration available."}

13. preview_generation · success · 23 ms
{"filesystem_paths_exposed": false, "format": "png"}

14. quantitative_spatial_fusion_eligibility · skipped · 0 ms
{"reason": "No geospatial co-registration available.", "scientific_classification": "UNVERIFIABLE"}

15. response_generation · success · 0 ms
{"language_model_used": false, "template_generated_summary": true}

16. sve_artifact_verification · success · 0 ms
{"checksum_verified": true}

17. sve_model_load · success · 3072 ms
{"device": "cpu", "reused": false}

18. sve_eligibility_check · success · 0 ms
{"input": "optical_rgb_like"}

19. sve_image_preparation · success · 0 ms
{"transform": "verified_openclip_validation_transform"}

20. sve_image_embedding · success · 178 ms
{"cache": "miss", "device": "cpu"}

21. sve_scene_priors · success · 3051 ms
{"calibrated": false, "top_labels": ["agricultural land", "urban or built-up area", "arable land", "residential area", "industrial area"]}

22. sve_routing_support · success · 0 ms
{"influence_applied": false, "support_state": "advisory"}

23. sve_eligibility_check · skipped · 0 ms
{"input": "raw_sar", "reason": "validated_generated_rgb_required"}

```

Scientific Limitations

- Optical and SAR percentages are deterministic candidate evidence, not calibrated semantic probabilities.
- Water-like and structural-likelihood evidence can be confused by shadow, terrain, speckle, and strong scatterers.
- Quantitative joint answering requires exact pixel and geospatial alignment; qualitative source comparison does not.
- This deterministic baseline is not a trained or calibrated land-cover classifier.

- Dark smooth terrain, cloud shadow, terrain shadow, and radar shadow can resemble water-like evidence.
- Bright terrain, speckle, vegetation structure, and isolated strong scatterers can resemble built-up or structural evidence.
- Visible-spectrum green dominance is only vegetation support; it is not NDVI and does not use NIR unless explicitly identified and implemented.
- Analysis may use a reduced grid capped at 1024 pixels on the longest side, so small features can be omitted.
- Adaptation used a geographically limited Lithuania subset.
- Adaptation imagery is summer-only.
- Version 1 supports RGB and scientifically approved RGB-like optical imagery only.
- Evidence is scene-level and does not localize pixels or objects.
- Similarity scores are not calibrated probabilities and are not ground truth.
- Validation retrieval improved over generic OpenCLIP on the recorded split but remains low in absolute terms.
- Outputs are not ground truth and must be reviewed with source imagery and mission context.
- Learned change-detector output is a model-generated binary prediction and is not ground truth.
- No hidden geolocation, unsupported causal interpretation, or calibrated semantic probability is produced.
- SatQuery Vision Encoder v1 provides scene-level embedding evidence. It does not produce pixel-level segmentation, object grounding, calibrated probabilities, or ground truth.

Reproducibility Summary

Request Id: 81496f78-f1f2-470e-92b0-657cc4947325

Tool Version: satquery-sar-routing-1.1

Method Configuration: {"auto_detected_modality": "optical_rgb", "effective_modality": "optical_rgb", "has_primary_image": true, "has_secondary_image": true, "input_mode": "cross_modal", "primary_band_count": 3, "primary_image_modality": "optical_rgb", "primary_modality": "optical", "primary_representation": "display_preview", "query": "Where do both modalities agree?", "representation": "display_preview", "secondary_modality": "sar"}

Generated Artifact List: ["GeoVision_SatQuery_cross_modal_analysis_81496f78-f1f2-470e-92b0-657cc4947325.pdf", "GeoVision_SatQuery_cross_modal_analysis_81496f78-f1f2-470e-92b0-657cc4947325.json", "GeoVision_SatQuery_cross_modal_analysis_81496f78-f1f2-470e-92b0-657cc4947325.zip"]

Cache: {"cache_key_prefix": "43fc9f6d7b76", "cached": false, "original_generation_timestamp": "2026-09-05T12:58:35.918989Z", "retrieval_timestamp": "2026-09-05T12:58:35.918989Z", "tool_version": "satquery-sar-routing-1.1"}

Restart Behavior: Result and artifact caches are bounded, process-local, and cleared when the backend restarts.